

Surface Area and Volumes

Exercise = 10.3

Q1)- Find the Volume of the right circular cone with:

(ii) Radius 3.5 cm, height 12 cm.

Sol. $r = 3.5 \text{ cm}$, $h = 12 \text{ cm}$

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times (3.5)^2 \times 12$$

$$= \frac{22}{7} \times 12 \times 25 = \frac{22}{7} \times \left(\frac{35}{10}\right)^2$$

$$= \frac{22}{7} \times \frac{35}{10} = \frac{22}{7} \times 7 \times 7$$

$$= 22 \times 7$$

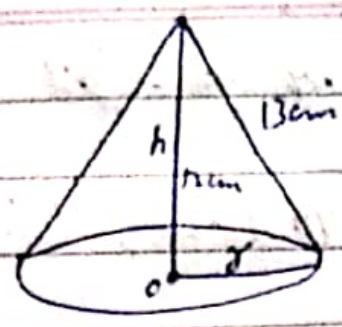
$$= 154 \text{ cm}^3.$$

Q2:- Find the capacity in litres of a conical vessel

with:

(i) Height 12 cm, slant height 13 cm.

Sol.



$$h = 12 \text{ cm}$$

$$l = 13 \text{ cm}$$

Since $l^2 = h^2 + r^2$

$$(13)^2 = (12)^2 + r^2$$

$$169 = 144 + r^2$$

$$\Rightarrow 169 - 144 = r^2$$

$$\Rightarrow r^2 = 25$$

$$\Rightarrow r = \sqrt{25}$$

$$\Rightarrow r = 5 \text{ cm}$$

$\therefore$ Volume of conical vessel.

$$= \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times (5)^2 \times 12$$

$$= \frac{22 \times 25 \times 4}{7} = \frac{2200 \text{ cm}^3}{7}$$

$$= \frac{2200}{7} \times \frac{1}{1000} \text{ l}$$

$$= \frac{22}{7} \times \frac{1}{10}$$

$$= \frac{11}{35} \text{ l.}$$

Q4: If the volume of a right circular cone of height 9cm is $48\pi \text{ cm}^3$, Find the diameter of its base.

Sol. $h = 9\text{cm}$
Volume = $48\pi \text{ cm}^3$

$$\Rightarrow \frac{1}{3}\pi r^2 h = 48\pi$$

$$\frac{1}{3}\pi r^2 \times 9 = 48\pi$$

$$3r^2 = 48$$

$$r^2 = \frac{48}{3}$$

$$r^2 = 16$$

$$r = \sqrt{16}$$

$$r = 4\text{cm}$$

$$\therefore \text{Diameter} = 2r$$

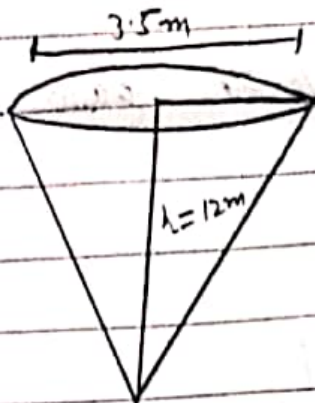
$$= 2 \times 4\text{cm}$$

$$= 8\text{cm}$$

Q5: A conical pit of top diameter 3.5m is 12m deep

What is its capacity in kilolitres?

Soln



$$r = \frac{D}{2} = \frac{3.5 \text{ m}}{2}$$

$$= 1.75$$

$$h = 12 \text{ m}$$

i. Capacity (Volume) of conical pit = $\frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times (1.75)^2 \times 12$$

$$= \frac{88}{7} \times \left(\frac{357}{100} \right)^2 \text{ m}^3$$

$$= \frac{88}{7} \times \left(\frac{7}{4} \right)^2 \text{ m}^3$$

$$= \frac{88}{7} \times \frac{7 \times 7}{4 \times 4} \text{ m}^3$$

$$= \frac{77 \text{ m}^3}{2} = 38.5 \text{ m}^3$$

$$= 38.5 \text{ kl.}$$

$$- 1000 \text{ cm}^3 = 1 \text{ l}$$

$$1000 \times 1 \text{ cm} = 1 \text{ l}$$

$$\frac{1 \text{ cm}^3}{1000} = 1 \text{ l}$$

$$- 1 \text{ kl} = 1000 \text{ l}$$

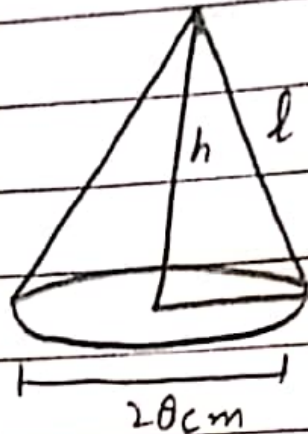
$$1000 \times 1 \text{ l} = 1 \text{ kl}$$

$$\frac{1 \text{ l}}{1000} = 1 \text{ kl}$$

$$- 1 \text{ m}^3 = 1000 \text{ l} / 1 \text{ kl}$$

Q6: The Volume of a right circular cone is 9856 m^3 .
If the diameter of the base is ~~28~~ 28 cm , find

(i) Height of the cone.



Sol Volume = 9856 cm^3

$$r = \frac{D}{2} = \frac{28}{2} \text{ cm}$$

$$r = 14 \text{ cm}$$

- Volume of rt. circular cone = 9856

$$\frac{1}{3} \pi r^2 h = 9856$$

$$\frac{1}{3} \times \frac{22}{7} \times (14)^2 \times h = 9856$$

$$\frac{1}{3} \times \frac{22}{7} \times 14 \times 14 \times h = 9856$$